

Amplify Analytix BA Case — Foundations

Start here: the fundamentals, the keywords, the dataset, and the formulas — explained from scratch

How to use this: read it top to bottom once. It assumes you know nothing about the case and builds the foundation so the numbers, words, and formulas all make sense — then you're ready for the deeper prep pack and a mock interview.

1 What a case-study interview is (and what this one tests)

A **case study** is a mini real-world problem the interviewer gives you to see how you think. You're not expected to have a perfect answer — they're watching *how* you break the problem down, handle messy data, make sensible choices, and explain them to non-technical people.

This case (for a **Business Analyst** role at **Amplify Analytix**, a data-analytics consultancy) has two parts:

- **Task 1 — Data analysis:** you get a spreadsheet of customers and are asked to analyse it and recommend what to do about the top customers. This tests your **data skills and business judgement**.
- **Task 2 — Case-study proposal:** a written business problem (different regions measure their sales 'funnel' differently) and you propose how you'd fix it. This tests your **structured thinking and consulting instincts** — no data needed.

2 The business context (so the data makes sense)

The data comes from **Henkel**, a large company that sells **adhesives and industrial products to other businesses** (this is **B2B** — business-to-business, not selling to you and me). So each 'customer' is actually another **company** that buys from Henkel.

Three teams inside Henkel care about these customers, and the whole analysis is really about helping them:

- **Sales** — wins deals and grows revenue. Cares about win rate and pipeline.
- **Marketing** — creates interest and nurtures customers (emails, campaigns). Cares about reach and engagement.
- **Customer Experience (CX)** — makes sure orders are delivered on time and customers stay happy. Cares about delivery scores.

The one idea to hold onto: every number in Task 1 is answering 'which customers matter most, and what should Sales, Marketing, or CX *do* about them?'

3 The keywords, in plain English (the foundation)

3.1 Money & growth

Term	In plain English
NES (Net External Sales)	The money a customer actually paid Henkel. 'Net' = after returns/credits; 'external' = real outside sales, not internal transfers. Simplest way to say it: revenue from that customer .
YoY growth	'Year over year.' This year's sales vs last year's, as a %. E.g. 110 vs 100 = +10%.
CAGR	Compound Annual Growth Rate — the smoothed average yearly growth over several years. It answers 'what steady % per year would take sales from the start value to the end value?'
Tenure / Recency / Purchase cycle	How long they've been a customer / how recently they last bought / how often they typically buy. Together they hint at loyalty and whether a customer is 'overdue.'

3.2 The sales pipeline (deals)

Term	In plain English
Opportunity	A potential deal — a specific chance to sell something to a customer. Tracked in the sales software (here called ACE , a CRM).
Won / Lost / Open / In-progress / Stopped	The status of an opportunity. Won = closed successfully; Lost = customer said no / bought elsewhere; Open / In-progress = still being worked; Stopped = abandoned.
Win rate	Of the deals that reached a decision (won or lost), what share did we win? A core measure of sales effectiveness. Below 50% at big accounts is a red flag.
Opportunity conversion rate	The share of opportunities that turned into sales. (In this dataset it's actually calculated as $\text{won} \div (\text{won} + \text{lost})$ — see the formulas.)
CRM	Customer Relationship Management software — the system that records every customer, contact and deal (e.g. Salesforce; here 'ACE').

3.3 Customer value & company info

Term	In plain English
Segment (Champions, Promising, Sustainers, Need Attention, Drifters)	Customers grouped by how valuable / healthy they are, using a machine-learning model. Champions = best; Drifters / Need Attention = weakest. It's a way to treat similar customers similarly.
Firmographics	'Demographics, but for companies' — facts about the customer's business, like its size, industry and overall revenue.
D&B (Dun & Bradstreet)	A company that sells business data. 'D&B revenue' / 'Global Ultimate revenue' = the total revenue of the customer's parent company — useful to spot when several customers belong to one big group.
SBU / Business unit (AMO, APC, ACM, AMI...)	Which part of Henkel's business the customer sits in — different product lines/divisions.

3.4 Delivery (Customer Experience)

Term	In plain English
POT-C	'On-time vs Communicated.' Did we deliver by the date we promised ? Measures how well our own process runs. ~1.0 = on time.
POT-AR	'On-time vs Actual Requested.' Did we deliver by the date the customer originally asked for ? Measures the customer's experience. If POT-C is high but POT-AR is low, we're hitting our own (later) promises but not what the customer really wanted.

3.5 Marketing & the funnel (needed for Task 2)

Term	In plain English
Emails sent / opened, engagement	How much marketing activity a customer has received and responded to. In this data it's mostly missing or zero — a coverage/recording gap.
Lead	A potential customer / contact who might buy one day.
Funnel (or pipeline)	The journey a lead takes from first contact to becoming a paying customer, split into stages that narrow like a funnel.
MQL (Marketing Qualified Lead)	A lead marketing judges interested/ready enough to hand to sales.
SQL (Sales Qualified Lead)	A lead sales has accepted as genuinely worth pursuing.
Known → MQL → SQL → Opportunity → Customer	The five-stage funnel Task 2 proposes as the shared language: a known contact → marketing-ready → sales-accepted → a real deal → a won customer.
Entry / exit rule, owner, timestamp	For each stage: what gets a lead in , what moves it on , who is accountable, and a date that can't be overwritten (so you can measure how long things take).

Term	In plain English
System of record	The one system treated as the 'source of truth' for a piece of data.

4 The dataset, explained simply

Open the **Data** sheet and picture a giant table: **one row = one customer company** (3,056 of them), and **53 columns** of facts about each. The columns fall into groups:

Column group	What's in it	Which team cares
Sales by year	Revenue (NES) for 2021, 2022, 2023, 2024, 2025, and Q1-2026; plus growth % and CAGR	Sales / Finance
Pipeline	Counts of opportunities: won, lost, open, in-progress, stopped; a conversion rate	Sales
Marketing	Emails sent / opened; a marketing rank (mostly missing)	Marketing
Firmographics	D&B parent-company revenue; firmographic rank	Sales / Strategy
Delivery (CX)	POT-C and POT-AR delivery scores	Customer Experience
Segmentation	The ML segment (Champions...Drifters) and business unit (SBU)	Everyone
Flags	'Missing data' markers for sales / marketing / firmographics	You (the analyst)

The other sheets are helpers: **Definitions** explains each metric; **Analysis_Approach** lists the assumptions and data-quality issues; **Analysis_EDA** and **Analysis_Top10** hold the live formulas; **Case_Study_QA** and **Slide_Talk_Track** are interview prep.

The most important thing about this dataset: it's messy on purpose. Lots of columns are missing (marketing 86% missing, delivery 87%), and some definitions don't match the actual numbers. Spotting and *stating* that — instead of trusting it blindly — is the biggest thing they're testing.

5 The formulas, with the intuition behind them

You don't need to memorise Excel syntax — you need to explain *what each number means and why*. Here's each one in words, with a worked example.

Total sales (Total NES 2021–25) — just add up a customer's five yearly sales figures.

$$\text{Total NES} = \text{NES}_{2021} + \text{NES}_{2022} + \text{NES}_{2023} + \text{NES}_{2024} + \text{NES}_{2025}$$

Example: €100m + €120m + €130m + €110m + €90m = €550m over five years.

Watch out: the ready-made 'nes' column secretly covers 2023–2026 (a partial year), so it isn't the true total — the analyst rebuilt it from the yearly columns.

Year-on-year (YoY) growth — how much this year changed versus last year.

$$\text{YoY growth} = (\text{this year} / \text{last year}) - 1$$

Example: 2025 = €90m, 2024 = €110m → 90/110 – 1 = –18% (down 18%).

CAGR (smoothed yearly growth) — the steady % per year that gets you from the first year to the last.

$$\text{CAGR} = (\text{end value} / \text{start value})^{(1 / \text{number of years})} - 1$$

Example: €100m in 2021 → €150m in 2025 (4 years): (150/100)^(1/4) – 1 = 1.5^0.25 – 1 ≈ +10.7% a year.

Watch out: 'number of years' is the gaps between years (2021→2025 = 4), not 5.

Q1-2026 annualised — turn one quarter into a rough full-year estimate to compare momentum.

$$\text{Annualised 2026} = \text{Q1-2026} \times 4$$

Example: Q1-2026 = €25m → ×4 = €100m implied for the year; compare that to 2025 to see the trend.

Watch out: it's only directional — one quarter isn't a real trend, so label it.

Customer concentration — how much of total sales sits with the biggest few customers.

$$\text{Concentration} = (\text{sales of top N customers}) / (\text{sales of all customers})$$

Example: Top 10 = €8.7bn, all customers = €48.4bn → 8.7 / 48.4 = 18% (10 names = 18% of the book).

Segment share — how much of total sales each customer segment contributes.

$\text{Segment share} = (\text{sales of that segment}) / (\text{sales of all segments})$

Example: Promising segment = €18bn of €48.4bn → 37% of sales.

Win rate — of the deals that were decided, how many did we win.

$\text{Win rate} = \text{Won} / (\text{Won} + \text{Lost})$ [open/in-progress deals excluded]

Example: 60 won, 40 lost → $60 / (60 + 40) = 60\%$.

Data completeness — what share of customers actually have data for a field.

$\% \text{ complete} = \text{customers with data} / \text{total customers}$

Example: 2,051 of 3,057 have sales history → 67% complete (so 33% is missing).

The clever guard: a delivery score of **0.00** is treated as 'no data,' not '0% on time.' Otherwise thousands of blank accounts would drag the average down and point the CX team at the wrong customers. In Excel:
`=IF(score=0,"no data", score).`

6 The story the numbers tell (say it like this)

Once you understand the pieces, Task 1 is really one short story:

- The business is **shrinking** — sales peaked in 2023 and have fallen since.
- It's **dangerously concentrated** — ~3% of customers make ~60% of revenue, and those big accounts are the ones declining fastest.
- We **lose more deals than we win** at the top accounts (win rate under 50%) — that's a **pricing/qualification** problem, not a lack of demand.
- **Marketing is barely present** — only ~12% of customers have ever been emailed, and the Promising segment (37% of sales) has had **zero** emails.
- And we can only half-trust the data, because so much is **missing** — so every recommendation is stated with that caveat.

7 Task 2 fundamentals (the funnel problem)

Task 2 has no data — it's a thinking exercise. The problem: a global company's regions all track the same sales funnel (Known → MQL → SQL → Opportunity), but each defines and counts the stages differently and uses different systems — so leadership can't compare regions or trust the numbers.

The whole answer in one line: *standardise the definition, not the process.* Agree what each stage *means* (the entry rule, the owner, the date), but let each region keep *how* they actually sell. That's what makes it acceptable to the regions instead of a fight.

Three ideas make it work: **core + local** (regions can add small steps inside a stage but can't invent new ones), **ownership** (one named owner per stage + a change process, or it drifts apart again), and **rules for messy reality** (leads skip stages and go backwards, so count skips as passed and log reversals instead of erasing dates). The killer evidence: the pack itself defines the funnel two different ways — proof the project is needed.

8 How to use this to prepare

- **First**, be able to say what each keyword in §3 means in one sentence, without notes.
- **Then**, be able to explain each formula in §5 in words (not Excel) and why it's done that way.
- **Then**, tell the §6 story out loud in 60 seconds.
- **Finally**, for Task 2, be able to say 'standardise the definition, not the process' and explain the five stages.

Quick self-test (answers are all in this guide): What does POT-AR measure, and how does it differ from POT-C? Why is 18% concentration a risk here, not a strength? Why is a sub-50% win rate a pricing problem and not a demand problem? What's the difference between an MQL and an SQL? Why exclude 2026 from the rankings?

Next: once these fundamentals feel solid, move to the **Explainer & Q&A prep pack** (the deeper deck), and then do a mock interview. You'll be ready.